

JIATENG LIU

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Homepage: <https://lumos-jiateng.github.io/>

Google scholar: <https://scholar.google.com/citations?user=a5PfQQAAAAJ&hl=zh-CN>

RESEARCH INTEREST

Natural Language Processing, Computer Vision, Deep Learning

Multi-modal world knowledge representation learning, Knowledge + LLMs & VLMs, (Multi-modal) LLM agents.

RESEARCH STATEMENT

My research centers on developing adaptable representations of world knowledge to support human-like intelligence and reasoning. I focus on building agentic systems that continually learn from real-world signals, advancing fine-grained multimodal alignment and interaction. A key aspect of my work is knowledge management, ensuring models remain consistent, current, and trustworthy for rigorous, real-world applications where safety is paramount.

EDUCATION

University of Illinois Urbana Champaign

Ph.D in Computer Science

Sep 2025 - Present

Instructor: Prof. Heng Ji

University of Illinois Urbana Champaign

Master of Science (M.S.) in Computer Science

August 2023 - May 2025

Instructor: Prof. Heng Ji

Zhejiang University

Bachelor of Science, Computer Science.

Overall GPA: 3.94/4.0

September 2019- June 2023

Instructor: Prof. Mingli Song

PUBLICATIONS

[1] Analyzing and Internalizing Complex Policy Documents for LLM Agents

Submitted to ICLR 2026

Jiateng Liu, Zhenhailong Wang, Xiaojiang Huang, Yingjie Li, Xing Fan, Chenlei Guo, Ruhi Sarikaya, Heng Ji

[2] EVEDIT: Event-based Knowledge Editing with Deductive Editing Boundaries

Accepted by EMNLP 2024 Main

Jiateng Liu*, Pengfei Yu*, Yuji Zhang, Sha Li, Zixuan Zhang, Heng Ji

[3] Rescorla Wagner Steering Under Mixed and Inappropriate Contexts

Accepted by EMNLP 2025 Main

Rushi Wang*, **Jiateng Liu***, Cheng Qian, Yifan Shen, Yanzhou Pan, Zhaozhuo Xu, Ahmed Abbasi, Heng Ji, Denghui Zhang

[4] A Language First Approach for Procedure Planning

Accepted by ACL 2023 Findings

Jiateng Liu, Sha Li*, Zhenhailong Wang, Manling Li, Heng Ji

- [5] **PropaInsight: Toward Deeper Understanding of Propaganda on Techniques, Appeals, and Intents**
Accepted by COLING 2025
Jiateng Liu*, Lin Ai, Gary Liu, Hui Zheng, Payam Karisani, May Fung, Preslav Nakov, Julia hirschberg, Heng Ji
- [6] **If LLM Is the Wizard, Then Code Is the Wand: A Survey on How Code Empowers Large Language Models to Serve as Intelligent Agents**
Accepted by ICLR 2024 Workshop
Ke Yang*, **Jiateng Liu***, John Wu, Chaoqi Yang, Yi R. Fung, Sha Li, Zixuan Huang, Xu Cao, Xingyao Wang, Yiquan Wang, Heng Ji, Chengxiang Zhai
- [7] **Multimodal Policy Internalization for Conversational Agents**
Submitted to ICLR 2026
Zhenhailong Wang, **Jiateng Liu**, Amin Fazel, Ritesh Sarkhel, Xing Fan, Xiang Li, Chenlei Guo, Heng Ji, Ruhi Sarikaya
- [8] **ISACL: Internal State Analyzer for Copyrighted Training Data Leakage**
Accepted by EMNLP 2025 Findings
Guangwei Zhang, Qisheng Su, **Jiateng Liu**, Cheng Qian, Yanzhou Pan, Yanjie Fu, Denghui Zhang
- [9] **MINT: Evaluating LLMs in Multi-turn Interaction with Tools and Language Feedback**
Accepted by ICLR 2024
Xingyao Wang*, Zihan Wang*, **Jiateng Liu**, Yangyi Chen, Lifan Yuan, Hao Peng, Heng Ji
- [10] **Automating Financial Statement Audits with Large Language Models**
Accepted by AAAI Workshop 2024
Rushi Wang*, **Jiateng Liu***, Weijie Zhao, Shenglan Li, Denghui Zhang
- [11] **Atomic Reasoning for Scientific Table Claim Verification**
Yuji Zhang, Qingyun Wang, Cheng Qian, Jiateng Liu, Chenkai Sun, Denghui Zhang, Tarek Abdelzaher, Chengxiang Zhai, Preslav Nakov, Heng Ji
- [12] **Knowledge Overshadowing Causes Amalgamated Hallucination in Language Models**
Accepted by ACL 2025 Findings
Yuji Zhang, Sha Li, **Jiateng Liu**, Pengfei Yu, Yi R. Fung, Jing Li, Manling Li, Heng Ji
- [13] **Video-LLMs with Temporal Visual Screening**
Zheyu Fan, Jiateng Liu, Yuji Zhang, Zihan Wang, Yi R Fung, Manling Li, Heng Ji
- [14] **SYNTHIA: Novel Concept Design with Affordance Composition**
Accepted by ACL 2025 Main
Hyeonjeong Ha, Xiaomeng Jin, Jeonghwan Kim, **Jiateng Liu**, Zhenhailong Wang, Khanh Duy Nguyen, Ansel Blume, Nanyun Peng, Kai-Wei Chang, Heng Ji
- [15] **CurveCloudNet: Processing Point Clouds with 1D Structure**
Accepted by CVPR 2024
Colton Stearns, Alex Fu, **Jiateng Liu**, Jeong Joon Park, Davis Rempe, Despoina Paschalidou, Leonidas Guibas

WORK EXPERIENCE

Applied Scientist Intern at Amazon

Mentored by: Xiaojiang Huang, Yingjie Li, Xing Fan

Seattle, WA

May 2025 - August 2025

- Worked on Internalization of Policy Documents for LLM Agents
- Inclined

Teaching assistant at UIUC

Siebel School of Computing and Data Science

Advised by: Prof. Margaret M. Fleck

Champaign, IL

August 2023 - December 2023

- Teaching CS440 (Artificial Intelligence), designing problem sets, tutoring students.

Research assistant at UIUC

Siebel School of Computing and Data Science

Advised by: Prof. Heng Ji

Champaign, IL

December 2023 - present

- **Project: ECOLE Visual Analytics**
- Design a new video representation for a better understanding of low-level object interactions.
- Designed a novel architecture to process videos and object-level representation.
- Apply tracklets using SAM2 to obtain motion tokens for objects.
- Collect new video instruction tuning datasets for learning actions.
- **Project: Semafor Open Characterization**
- With a focus on real-world propaganda usage, training LLMs to detect misinformation.
- Rooted in established social science works, provide a new framework representing propaganda.
- Design a partially controlled pipeline for generating synthetic data.
- Use Label Studio to manually annotate data, collaborated with Kitware.Inc.

RESEARCH INTERN EXPERIENCE

Summer Internship at Stanford University

3D Reconstruction from Curve Data

August 2022 - December 2022

Advisors: Prof. Leonidas Guibas and Prof. Yanchao Yang

- Focused on developing CurveNet, a novel approach for 3D reconstruction leveraging curve data to enhance geometric detail and accuracy in generated models. Collaborated with a team to integrate this technology with existing 3D imaging systems.

Summer Internship at University of Illinois Urbana-Champaign

Research on Large Language Model (LLM) Agents

June 2023 - November 2023

Advisor: Prof. Heng Ji

- Conducted research aimed at improving the interpretability and reliability of LLM agents in natural language processing tasks. Participated in the design and testing of model frameworks and contributed to two published papers.

Research Project at Zhejiang University

Multi-Model Representation Learning and Efficient Transformers

October 2021 - June 2022

Advisor: Prof. Mingli Song

- Engaged in developing advanced machine learning models that efficiently process and integrate multiple data types. Improved transformer architectures for better performance and lower computational costs